

Assignment 1 - MovieLens Dataset Exploration

Submitted by: Sruthilaya Umasankari Soma Shanmuga Sundaram

Q1: To find mean ratings by men aged above 25, per genre

The screenshot below shows the average ratings by men >25 age per genre.

Insight obtained - The genre "Film Noir" has gained more appreciation with ~4 rating than other genres with average of ~3.5 rating. Extending the same analysis to female >25 age group also show the same interest which shows a general liking towards this genre.

Men >25 age rating average per genre

Genre	
Film-Noir	4.117140
Documentary	3.950192
War	3.940634
Drama	3.812309
Crime	3.764249
Mystery	3.759347
Animation	3.721569
Western	3.708494
Musical	3.700242
Romance	3.659748
Thriller	3.644025
Comedy	3.565456
Action	3.554547
Adventure	3.538637
Sci-Fi	3.509693
Fantasy	3.490408
Children's	3.475314
Horror	3.241089

Name: Rating, dtype: float64

Female >25 age rating average per genre

Genres	
Film-Noir	4.064204
War	3.976948
Documentary	3.946574
Musical	3.879710
Drama	3.835724
Animation	3.803525
Mystery	3.801381
Romance	3.766936
Crime	3.753047
Western	3.692266
Children's	3.668804
Thriller	3.659867
Comedy	3.658881
Adventure	3.591242
Fantasy	3.580167
Action	3.559380
Sci-Fi	3.503287
Horror	3.295800

Name: Rating, dtype: float64

Q2: Top 5 movies which is most reviewed (ie, has more people rating it)

The below screenshots show the top 5 movies which are most reviewed by people.

Insight Obtained - The most rated movie is American beauty which is the most hyped movie of that period with its oscar wins. Then it is followed by a franchise of Star Wars movies which has a collective 3x more ratings combined showing it has been one strong movie franchise.

```
... Top 5 Movies by Rating Count:
```

	Title	RatingCount
0	American Beauty (1999)	3428
1	Star Wars: Episode IV - A New Hope (1977)	2991
2	Star Wars: Episode V - The Empire Strikes Back...	2990
3	Star Wars: Episode VI - Return of the Jedi (1983)	2883
4	Jurassic Park (1993)	2672

```
Title
```

```
Star Wars: Episode VI - Return of the Jedi (1983)    2000-10-02 04:14:25.911550464
Star Wars: Episode V - The Empire Strikes Back (1980) 2000-10-04 22:26:35.886956416
Star Wars: Episode IV - A New Hope (1977)           2000-10-05 14:04:10.400200576
American Beauty (1999)                             2000-10-06 23:41:25.196324352
Jurassic Park (1993)                                2000-10-10 03:20:55.602919168
Name: Date, dtype: datetime64[ns]
```

Q3. Finding Average Rating by Age Groups

Here the age groups weren't specified directly on the dataset I obtained from the github since it could be preprocessed, with my assumption that 1 represents ≤ 18 , 18,25 represents 19-35, 35,45 represents 31-50, and 50,56 represents 51-70 age groups respectively I proceeded with this task.

Insight Obtained: The screenshot below shows the older the age group is the more higher rating they offer. This suggests older age groups know their liking and carefully pick movies closer to what they like therefore results in higher rating.

Age group ≥ 70 is missing because the dataset didn't contain any rating of users above 56 age.

```
AgeGroup
1. <=18      3.549520
2. 19-30     3.533299
3. 31-50     3.624050
4. 51-70     3.732677
Name: Rating, dtype: float64

print("Users with age code above 56:", (users['Age'] > 56).sum())
✓ 0.0s

Users with age code above 56: 0
```

Q4. Average rating of a Movie of my choice vs all movies the same year by the age groups 1. ≤ 18 2. ≥ 19 and ≤ 45 3. > 45

I have chosen Toy Story 2 (1999) for my analysis. Looked at all 1999 movies and broke down unique movies rated by 3 age buckets. Extended this to also compare how Toy Story 2 specifically performed vs the 1999 average it was rated above average by every single age group which was cool to see.

Insight obtained: The analysis showed Toy Story 2 is a rare film where the "kids movie" label is misleading, data shows adults were both the biggest audience and the biggest fans. Toy story should be one of the biggest hits since it has an average ~4 rating compared to all other movies from the year which has an average rating of ~3.5.

```
... Total movies from 1999: 283
```

	MovieID	Title	Genres
2166	2235	One Man's Hero (1999)	Drama War
2367	2436	Tea with Mussolini (1999)	Comedy
2376	2445	At First Sight (1999)	Drama
2377	2446	In Dreams (1999)	Thriller
2378	2447	Varsity Blues (1999)	Comedy Drama

```
AgeGroup_Q4
a. Under 18    221
b. 19 to 45   270
c. Above 45   253
Name: MovieID, dtype: int64
```

```
toystory2 = ratings[ratings['MovieID'] == 3114]
toystory2 = toystory2.merge(users[['UserID', 'AgeGroup_Q4']], on='UserID')

# count of number of ratings and average rating for Toy Story 2 by age group
toystory2_analysis = toystory2.groupby('AgeGroup_Q4')['Rating'].agg(['count', 'mean'])
toystory2_analysis.columns = ['NumRatings', 'AvgRating']
print(toystory2_analysis)
```

```
✓ 0.0s
```

	NumRatings	AvgRating
AgeGroup_Q4		
a. Under 18	99	4.202020
b. 19 to 45	1251	4.231015
c. Above 45	235	4.161702

```
#Identifying the average rating for all 1999 movies by age group
avg_1999 = merged_rating_1999.groupby('AgeGroup_Q4')['Rating'].mean()

# Adding the comparison column
toystory2_analysis['Avg_1999_Movies'] = avg_1999
toystory2_analysis['AboveAvg?'] = toystory2_analysis['AvgRating'] > toystory2_analysis['Avg_1999_Movies']
print(toystory2_analysis)
```

```
✓ 0.0s
```

	NumRatings	AvgRating	Avg_1999_Movies	AboveAverage?	AboveAvg?
AgeGroup_Q4					
a. Under 18	99	4.202020	3.525049	True	True
b. 19 to 45	1251	4.231015	3.486012	True	True
c. Above 45	235	4.161702	3.542485	True	True

Q5. Write a function that takes user_id + movie_id and returns all movies rated the same way

Wrote a function that takes user_id and movie_id, finds what rating that user gave the movie, then returns all other movies they rated the same way. Tested it with User 1 and Toy Story 2 and also tested the edge case where a user hasn't rated the movie at all.

```
... TEST 1 - Valid user and movie:
User 1 rated this movie: 4 stars

    Title      Rating
0      Erin Brockovich (2000)      4
1  Snow White and the Seven Dwarfs (1937)      4
2      Wizard of Oz, The (1939)      4
3          Gigi (1958)      4
4      Miracle on 34th Street (1947)      4
5      Ferris Bueller's Day Off (1986)      4
6      Airplane! (1980)      4
7          Bambi (1942)      4
8          Big (1988)      4
9      E.T. the Extra-Terrestrial (1982)      4
10         Titanic (1997)      4
11         Ponette (1996)      4
12         Antz (1998)      4
13      Girl, Interrupted (1999)      4
14         Hercules (1997)      4
15         Aladdin (1992)      4
16         Mulan (1998)      4
17      Hunchback of Notre Dame, The (1996)      4
18         Sixth Sense, The (1999)      4
19      Driving Miss Daisy (1989)      4
20      Run Lola Run (Lola rennt) (1998)      4
21  Star Wars: Episode IV - A New Hope (1977)      4
...
25      Dead Poets Society (1989)      4

TEST 2 - User who hasn't rated Toy Story 2:
User 999 has not rated MovieID 3114
```

Q6 Further Interesting Insights from my Exploration

1. Prolific users (users who rate 500+ movies) vs casual users (who rated <100 movies) - users who rate 500+ movies are almost 0.4 stars harsher on average than casual raters. This suggests the more you watch, more critical you get.

```
... RaterType
Casual (<100)      3095
Moderate (100-500) 2546
Prolific (500+)    399
Name: count, dtype: int64
```

```
merged_user_types = ratings.merge(user_types, on='UserID')
user_types_ratings = merged_user_types.groupby('RaterType')['Rating'].agg(['mean', 'count'])
user_types_ratings.columns = ['AvgRating', 'TotalRatings']
print(user_types_ratings.sort_values('AvgRating', ascending=False))
```

```
[100] ✓ 0.1s
```

```
... AvgRating TotalRatings
RaterType
Casual (<100)      3.774183      152907
Moderate (100-500) 3.630916      556993
Prolific (500+)    3.385424      290309
```

2. Hidden gems - used data-driven thresholds (25th percentile to median rating count) to find underseen but highly rated movies. Top result was Kurosawa's Sanjuro (4.61 avg) which most people have probably never heard of.

Total hidden gems found: 46			
	MovieID	NumRatings	AvgRating \
30	2905	69	4.608696
9	670	56	4.410714
8	668	47	4.404255
3	326	61	4.344262
41	3470	92	4.326087
25	2360	117	4.307692
34	2937	104	4.288462
42	3679	62	4.274194
4	363	62	4.258065
32	2925	57	4.228070

	Title	Genres
30	Sanjuro (1962)	Action Adventure
9	World of Apu, The (Apu Sansar) (1959)	Drama
8	Pather Panchali (1955)	Drama
3	To Live (Huozhe) (1994)	Drama
41	Dersu Uzala (1974)	Adventure Drama
25	Celebration, The (Festen) (1998)	Drama
34	Palm Beach Story, The (1942)	Comedy
42	Decline of Western Civilization, The (1981)	Documentary
4	Wonderful, Horrible Life of Leni Riefenstahl, ...	Documentary
32	Conformist, The (Il Conformista) (1970)	Drama

- Genre vs Age Group heatmap - visualized average rating for every genre x age group combination. Film-Noir is loved by everyone, Horror is disliked by everyone, and older audiences appreciate Animation way more than younger ones (nostalgia could be a factor).

Average Movie Rating by Genre and Age Group

